

# stat\_transported\_late\_strength\_frontier

*Specialization:* v1

**TL;DR.** The transported complier effect is forced into  $\Theta = [-1, 1]$ , and its honest expected-length frontier is governed by the effective strength  $t_n = n\mu_n^2/\kappa_n$ . Conditional on each admissible deterministic source-covariate, transport-weight, and propensity geometry, a converse and one oracle source-score inversion sequence match at order  $\min\{1, t_0^{-1/2}\}$  for every fixed threshold  $t_0$ , even when  $\mu_n \rightarrow 0$  and  $\kappa_n \rightarrow \infty$ . In the separate uniform finite-cell experiment, target cell probabilities are learned from the covariate-only target sample with no additional minimax-order loss; the feasible conclusion also extends to known regular nonuniform source cells and known cell-varying propensities.

## 1 Project justification

**Gap.** Chen and Huang (2025, [1]) give the transported-CACE identifying ratio and regular weighted or multiply robust asymptotics; Zeng et al. (2024, [10]) give regular transport/generalization efficiency and minimax theory for target causal effects. Choi and Shen (2019, [6]) address weak-IV confidence sets in a linear two-sample IV model, while Mikusheva (2010, [7]) analyzes a different invariant linear-IV experiment. The accepted-bank discrete-ATE minimax result studies MSE under unconfoundedness with dimension and overlap, not an IV-ratio confidence-length problem. These comparators do not give a fixed-geometry honest expected-length frontier for a bounded transported causal ratio with  $\mu_n \rightarrow 0$  and  $\kappa_n \rightarrow \infty$ , nor a no-order-loss theorem for learning target cell weights from a covariate-only sample.

**Niche.** The note fixes the deterministic geometry  $(P_S^X, w, e)$  before taking the conditional minimax risk, isolates the index  $t_n = n\mu_n^2/\kappa_n$ , and then separately studies target-weight learning where uniform or regular source cell probabilities and the assignment propensity are known.

**Fill.** The paper proves matching lower and upper oracle frontiers, including a converse conditional on every admissible fixed geometry, and proves that estimating target cell probabilities causes no order loss on the uniform finite-cell class or its known regular nonuniform extension. This is not a general learned-nuisance result: unknown assignment propensities and feasibility beyond the stated finite-cell designs remain outside the established scope.

## 2 Related work

Chen and Huang (2025, [1]) provide the transported-CACE Wald identifying ratio and regular weighted and multiply robust asymptotics. Their guarantee is not a conditional minimax bound for honest confidence-set length, does not fix an arbitrary deterministic  $(P_S^X, w, e)$  geometry, and does not cover the joint boundary  $\mu_n \rightarrow 0, \kappa_n \rightarrow \infty$ . Zeng, Kennedy, Bodnar, and Naimi (2024, [10]) provide regular efficiency and minimax theory for transported target causal effects; that is a different estimand and regularity regime from a bounded transported complier ratio with a vanishing first stage. Rudolph and van der Laan (2017, [2]) likewise develop regular semiparametric transported-encouragement inference under a different observed-data design. Mikusheva (2010, [7]), Ma (2023, [3]), and Smucler, Lanni, and Masip (2025, [4]) concern weak-IV-robust inference in one-population or invariant linear-IV settings rather than a bounded transported CACE with diverging weight dispersion. Choi and Shen (2019, [6]) give weak-instrument-robust confidence sets for linear two-sample IV, but their experiment does not have a covariate-only target sample, transport density ratio, or conditional minimax expected-length frontier; Choi, Gu, and Shen (2018, [5]) has the same relevant linear-two-sample contrast. Within the accepted CausalSmith bank, `stat_discrete_ate_minimax_loggap` is also distinct: it proves a discrete unconfounded-ATE MSE frontier driven by dimension and overlap, rather than IV-ratio identification, transport/Kish geometry, or honest confidence-set length. The present results instead establish matched honest expected-length orders, first conditionally on each fixed transport geometry and then for the stated structured target-weight-learning experiments.

## 3 Setup

Target (estimand / structural parameter):  $\theta_T = \mathbb{E}_{P_n^F}[Y(1) - Y(0) \mid D(1) > D(0), S = 0]$ . Primary functional / estimator / diagnostic:

$$\theta_T = \mu_{Y,n}/\mu_n = \mathbb{E}_S[w(X)\Delta_Y(X)]/\mathbb{E}_S[w(X)\Delta_D(X)], \quad t_n = n\mu_n^2/\kappa_n$$

- $n$  — positive integer sequence;  $\mathbb{N}$ ; asymptotic index

*Definition.* Study-sample size with  $n \rightarrow \infty$ .

- $\mathcal{X}$  — measurable space;  $(\mathcal{X}, \mathcal{A})$ ; covariate support

*Definition.* Common covariate space.

- $X$  — random variable;  $\mathcal{X}$ ; covariate

*Definition.* Covariate under the population specified by context.

- $S$  — random variable;  $\{0, 1\}$ ; population label

*Definition.* Population indicator;  $S = 1$  is source and  $S = 0$  is target.

- $D(z)$  — binary potential random variable;  $\{0, 1\}$ ; receipt potential outcome

*Definition.* Treatment receipt under encouragement  $z \in \{0, 1\}$ .

- $Y(d)$  — bounded potential random variable;  $[0, 1]$ ; outcome potential outcome

*Definition.* Outcome under treatment  $d \in \{0, 1\}$ .

- $Y^Z(z)$  — assignment potential random variable;  $[0, 1]$ ; assignment outcome

*Definition.*  $Y^Z(z) = Y(D(z))$ .

- $P_n^F$  — full-data triangular-array probability law;  $\mathcal{P}(\{0, 1\} \times \mathcal{X} \times \{0, 1\}^2 \times [0, 1]^2)$ ; causal full-data law

*Definition.* Law of  $(S, X, D(0), D(1), Y(0), Y(1))$  at index  $n$ .

- $Z$  — binary random variable;  $\{0, 1\}$ ; instrument

*Definition.* Source encouragement assignment.

- $D$  — binary random variable;  $\{0, 1\}$ ; observed treatment

*Definition.* Observed treatment receipt in the source sample.

- $Y$  — bounded random variable;  $[0, 1]$ ; observed outcome

*Definition.* Observed outcome in the source sample.

- $\epsilon$  — real constant;  $(0, 1/2)$ ; overlap constant

*Definition.* Instrument-positivity constant.

- $c$  — real constant;  $(0, \infty)$ ; sampling ratio

*Definition.* Target-to-source sample-size limit.

- $N_n$  — positive integer sequence;  $\mathbb{N}$ ; target sample size

*Definition.* Target covariate-sample size satisfying  $N_n/n \rightarrow c$ .

- $O_i^S$  — random vector;  $\mathcal{X} \times \{0, 1\}^2 \times [0, 1]$ ; source observation

*Definition.*  $O_i^S = (X_i, Z_i, D_i, Y_i)$ .

- $X_j^T$  — random variable;  $\mathcal{X}$ ; target observation

*Definition.* Target covariate observation  $j$ .

- `P_S` — probability law;  $\mathcal{P}(\mathcal{X} \times \{0, 1\}^2 \times [0, 1])$ ; source observed-data law  
*Definition.* Observed source law induced by  $P_n^F$  given  $S = 1$  and source assignment.
- `P_S^X` — probability law;  $\mathcal{P}(\mathcal{X})$ ; source covariate law  
*Definition.*  $X$ -marginal of  $P_S$ .
- `P_T` — probability law;  $\mathcal{P}(\mathcal{X})$ ; target covariate law  
*Definition.*  $X$ -marginal of  $P_n^F$  conditional on  $S = 0$ .
- `e` — measurable function;  $\mathcal{X} \rightarrow [\varepsilon, 1 - \varepsilon]$ ; source propensity  
*Definition.*  $e(x) = P_S(Z = 1 \mid X = x)$ .
- `P` — triangular-array data law;  $(P_n^F, e)_{n \geq 1}$ ; data-generating sequence  
*Definition.* Joint law inducing the two observed samples.
- `w` — measurable density-ratio function;  $\mathcal{X} \rightarrow [0, \infty)$ ; transport weight  
*Definition.*  $w = dP_T/dP_S^X$ .
- `\Delta_Y` — measurable function;  $\mathcal{X} \rightarrow [-1, 1]$ ; source outcome ITT  
*Definition.*  $\Delta_Y(x) = \mathbb{E}_S[Y \mid Z = 1, X = x] - \mathbb{E}_S[Y \mid Z = 0, X = x]$ .
- `\Delta_D` — measurable function;  $\mathcal{X} \rightarrow [0, 1]$ ; source receipt ITT  
*Definition.*  $\Delta_D(x) = \mathbb{E}_S[D \mid Z = 1, X = x] - \mathbb{E}_S[D \mid Z = 0, X = x]$ .
- `\mu_{Y,n}` — real sequence;  $[-1, 1]$ ; transported outcome ITT  
*Definition.*  $\mu_{Y,n} = \mathbb{E}_S[w(X)\Delta_Y(X)]$ .
- `\mu_n` — positive real sequence;  $(0, 1]$ ; transported first stage  
*Definition.*  $\mu_n = \mathbb{E}_S[w(X)\Delta_D(X)]$ .
- `\kappa_n` — real sequence;  $[1, \infty)$ ; Kish dispersion  
*Definition.*  $\kappa_n = \mathbb{E}_S[w(X)^2]$ .
- `k_n` — positive integer sequence;  $\mathbb{N}$ ; overlap envelope  
*Definition.* Known envelope with  $k_n \rightarrow \infty$  and  $k_n = o(n^{1/2})$ .
- `t_n` — positive real sequence;  $(0, \infty)$ ; effective identification strength  
*Definition.*  $t_n = n\mu_n^2/\kappa_n$ .
- `\Theta` — compact parameter space;  $\mathbb{R}$ ; forced causal range  
*Definition.*  $\Theta = [-1, 1]$ .
- `\theta_T` — causal parameter;  $\Theta$ ; target CACE  
*Definition.*  $\theta_T = \mathbb{E}_{P_n^F}[Y(1) - Y(0) \mid D(1) > D(0), S = 0]$ .
- `\alpha` — real constant;  $(0, 1)$ ; coverage level  
*Definition.* Noncoverage level.
- `C_n` — random set;  $2^\Theta$ ; procedure  
*Definition.* Confidence set measurable with respect to its declared inputs.
- `\lambda` — measure;  $\mathcal{B}(\Theta) \rightarrow [0, 2]$ ; set length  
*Definition.* Lebesgue length restricted to  $\Theta$ .

- $t_0$  — positive real number;  $(0, \infty)$ ; frontier argument

*Definition.* Lower effective-strength threshold.

- $c_0$  — real constant;  $(0, \infty)$ ; converse constant

*Definition.* Lower-frontier constant.

- $C_0$  — real constant;  $(0, \infty)$ ; attainment constant

*Definition.* Upper-frontier constant.

- $t_c$  — real constant;  $(0, \infty)$ ; phase threshold

*Definition.* Constant-order transition threshold.

- $\rho$  — real constant;  $(0, 1/4)$ ; fixed-geometry Le Cam calibration

*Definition.*  $\rho = (1 - \alpha)/8$  is the continuum perturbation calibration.

- $\mathcal{P}_n$  — class of triangular-array laws;  $2^{\mathcal{P}}$ ; model class

*Definition.* Main transported-IV model class.

- $\mathcal{N}_n$  — class of triangular-array laws;  $2^{\mathcal{P}_n}$ ; feasible submodel

*Definition.* Growing finite-cell subclass.

- $\mathcal{C}^{\{\text{or}\}}$  — class of procedure sequences;  $2^{\{(C_n)_{n \geq 1}\}}$ ; oracle procedures

*Definition.* Oracle-honest procedure class.

- $\mathcal{C}^{\{\text{cell}\}}$  — class of procedure sequences;  $2^{\{(C_n)_{n \geq 1}\}}$ ; feasible procedures

*Definition.* Finite-cell honest procedure class without  $w$ .

- $R$  — risk functional;  $[0, 2]$ ; frontier risk

*Definition.* Worst-case limiting expected set length.

- $V^*$  — value function;  $(0, \infty) \rightarrow [0, 2]$ ; main value

*Definition.* Oracle minimax frontier.

- $V_{\mathcal{N}}^*$  — value function;  $(0, \infty) \rightarrow [0, 2]$ ; submodel value

*Definition.* Finite-cell oracle minimax frontier.

- $\mathcal{G}$  — admissible deterministic geometry array;  $\mathcal{G}$ ; fixed geometry

*Definition.* A sequence  $(P_S^X, w, e)_{n \geq 1}$  satisfying `def:admissible-geometry-class`.

- $\mathcal{G}$  — class of deterministic geometry arrays;  $2^{\{(P_S^X, w, e)_{n \geq 1}\}}$ ; geometry class

*Definition.* The class defined in `def:admissible-geometry-class`.

- $\mathcal{P}_n(\mathcal{G})$  — fixed-geometry model slice;  $2^{\mathcal{P}_n}$ ; conditional model class

*Definition.* Members of  $\mathcal{P}_n$  whose source covariate law, transport weight, and propensity equal the geometry  $\mathbf{g}$ .

- $\mathcal{C}^{\{\text{or}\}}(\mathcal{G})$  — class of conditionally oracle-honest sequences;  $2^{\{(C_n)_{n \geq 1}\}}$ ; conditional procedures

*Definition.* Oracle procedures honest uniformly over  $\mathcal{P}_n(\mathbf{g})$ .

- $V_{\mathcal{G}}^*$  — fixed-geometry value function;  $(0, \infty) \rightarrow [0, 2]$ ; conditional frontier

*Definition.* The conditional minimax expected-length frontier defined in `def:fixed-geometry-value`.

- $q_{\cdot\{x,n\}}$  — positive source-cell probability array;  $(0,1]$ ; known source cell probability

*Definition.*  $q_{x,n} = P_S^X(X = x)$  for  $x \in \{1, \dots, k_n\}$ .

- $c_{-}$  — positive real constant;  $(0,1]$ ; cell regularity constant

*Definition.* Lower regularity constant for source-cell probabilities.

- $c_{+}$  — real constant;  $[1, \infty)$ ; cell regularity constant

*Definition.* Upper regularity constant for source-cell probabilities.

- $\mathcal{N}_n^{\{\mathrm{reg}\}}$  — class of triangular-array laws;  $2^{P_n}$ ; widened feasible submodel

*Definition.* The regular nonuniform finite-cell class in `def:regular-finite-cell-class`.

- $\mathcal{C}^{\{\mathrm{reg}\}}$  — class of regular-cell honest procedure sequences;  $2^{\{(C_n)_{n \geq 1}\}}$ ; widened feasible procedures

*Definition.* Procedures using the two samples and known  $(q_{x,n})$  and  $e$ , but not  $w$ , that are honest over  $\mathcal{N}_n^{\mathrm{reg}}$ .

## 4 Assumptions

**Assumption 1** (ass:full-data-support).  $P_n^F\{S \in \{0,1\}, D(0), D(1) \in \{0,1\}, Y(0), Y(1) \in [0,1]\} = 1$ .

Novel. *Bounded potential outcomes are routine for minimax expected-length analysis of binary or rescaled outcomes. This load-bearing restriction forces the causal ratio to have a finite range rather than importing an arbitrary truncation.*

**Assumption 2** (ass:population-presence).  $P_n^F(S = s) > 0$  for  $s \in \{0,1\}$ .

Standard (*two-population support*, [1]).

**Assumption 3** (ass:two-sample-array).  $(O_i^S)_{i=1}^n \sim P_S^{\otimes n}$ ,  $(X_j^T)_{j=1}^{N_n} \sim P_T^{\otimes N_n}$ ,  $(O_i^S)_{i=1}^n \perp (X_j^T)_{j=1}^{N_n}$ , and  $N_n/n \rightarrow c$ .

Standard (*independent two-sample triangular-array sampling*, [1]).

**Assumption 4** (ass:instrument-overlap).  $\varepsilon \leq e(X) \leq 1 - \varepsilon$   $P_S$ -almost surely.

Standard (*instrument positivity*, [1]).

**Assumption 5** (ass:source-observation).  $P(Z = 1 \mid X, S = 1) = e(X)$ ,  $D = D(Z)$ , and  $Y = Y(D)$  conditional on  $S = 1$ .

Standard (*source assignment and consistency*, [1]).

**Assumption 6** (ass:iv-randomization).  $Z \perp (Y(0), Y(1), D(0), D(1)) \mid (X, S = 1)$ .

Standard (*conditional instrument exchangeability*, [1]).

**Assumption 7** (ass:iv-exclusion).  $Y^Z(z) = Y(D(z))$   $P_n^F$ -almost surely conditional on  $S = s$ , for  $z, s \in \{0,1\}$ .

Standard (*instrument exclusion restriction*, [1]).

**Assumption 8** (ass:iv-monotonicity).  $D(1) \geq D(0)$   $P_n^F$ -almost surely conditional on  $S = s$ , for  $s \in \{0,1\}$ .

Standard (*no-defiers monotonicity*, [1]).

**Assumption 9** (ass:outcome-transport).  $\mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 0] = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1]$  for  $P_T$ -almost every  $x$ .

Standard (*conditional transport of the assignment-outcome contrast*, [1]).

**Assumption 10** (ass:receipt-transport).  $\mathbb{E}_{X \mid S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 0]] = \mathbb{E}_{X \mid S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 1]]$ .

Standard (*integrated transport of the first stage*, [1]).

**Assumption 11** (ass:target-complier-positivity).  $P_n^F(D(1) > D(0) \mid S = 0) > 0$ .

Standard (*positive target complier share*, [1]).

**Assumption 12** (ass:transport-domination).  $P_T \ll P_S^X$  on  $\mathcal{X}$ .

Standard (transport covariate-support inclusion, [1]).

**Assumption 13** (ass:weight-envelope).  $0 \leq w(X) \leq 2k_n$   $P_S$ -almost surely.

Novel. Calibrated or coarsened transport designs can impose an externally auditable density-ratio cap while allowing the cap to grow with increasingly distinct target populations.

**Assumption 14** (ass:weight-second-moment).  $\kappa_n \leq k_n$ .

Novel. The condition permits growing Kish design effects realized by finite calibration tables while retaining a transparent overlap scale.

**Assumption 15** (ass:degrading-array).  $k_n \rightarrow \infty$ ,  $k_n = o(n^{1/2})$ , and  $\mu_n \rightarrow 0$ .

Novel. A target recruitment shift can simultaneously concentrate covariates in poorly represented strata and reduce the target complier share.

**Assumption 16** (ass:finite-cell-source).  $\mathcal{X} = \{1, \dots, k_n\}$ ,  $P_S^X$  is uniform, and  $e(X) = 1/2$   $P_S$ -almost surely.

Novel. Published calibration tables can fix source-cell shares and assignment probabilities while a registry leaves only the target cell probabilities unknown.

## 5 Definitions

**Definition 1** (def:model-class). Defined object.  $\mathcal{P}_n$

Construction.

$\mathcal{P}_n = \{P = (P_n^e, e) : P \text{ satisfies } \text{ass:full-data-support, ass:population-presence, ass:two-sample-array, ass:instrument-overlap, ass:source-observation, ass:iv-randomization, ass:iv-exclusion, ass:iv-monotonicity, ass:outcome-transport, ass:receipt-transport, ass:target-complier-positivity, ass:transport-domination, ass:weight-envelope, ass:weight-second-moment, and ass:degrading-array at index } n\}$

Member properties. *ass:full-data-support; ass:population-presence; ass:two-sample-array; ass:instrument-overlap; ass:source-observation; ass:iv-randomization; ass:iv-exclusion; ass:iv-monotonicity; ass:outcome-transport; ass:receipt-transport; ass:target-complier-positivity; ass:transport-domination; ass:weight-envelope; ass:weight-second-moment; ass:degrading-array.*

**Definition 2** (def:finite-cell-class). Defined object.  $\mathcal{N}_n$

Construction.  $\mathcal{N}_n = \{P \in \mathcal{P}_n : P \text{ satisfies } \text{ass:finite-cell-source at index } n\}$ .

Member properties. *ass:full-data-support; ass:population-presence; ass:two-sample-array; ass:instrument-overlap; ass:source-observation; ass:iv-randomization; ass:iv-exclusion; ass:iv-monotonicity; ass:outcome-transport; ass:receipt-transport; ass:target-complier-positivity; ass:transport-domination; ass:weight-envelope; ass:weight-second-moment; ass:degrading-array; ass:finite-cell-source.*

**Definition 3** (def:oracle-honesty). Defined object. *oracle-honest sequence*

Construction.

$\mathfrak{C}^{\text{or}} = \{(C_n)_{n \geq 1} : C_n = C_n((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e) \subseteq \Theta, \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n} P(\theta_T \in C_n) \geq 1 - \alpha\}$

Inputs.  $O_i^S; X_j^T; w; e$ .

**Definition 4** (def:frontier-risk). Defined object.  $R((C_n), t_0)$

Construction.  $R((C_n), t_0) = \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)]$ .

**Definition 5** (def:oracle-value). Defined object.  $V^*(t_0)$

Construction.  $V^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}} R((C_n), t_0)$ .

**Definition 6** (def:finite-cell-oracle-value). Defined object.  $V_{\mathcal{N}}^*(t_0)$

Construction.

$$V_{\mathcal{N}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)]$$

**Definition 7** (def:finite-cell-feasible-honesty). Defined object. *weight-unknown finite-cell-honest sequence*

Construction.

$$\mathfrak{C}^{\text{cell}} = \{(C_n)_{n \geq 1} : C_n = C_n((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}) \subseteq \Theta, \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n} P(\theta_T \in C_n) \geq 1 - \alpha\}$$

Inputs.  $\mathcal{O}_i \sim \mathcal{S}; X_j \sim T$ .

**Definition 8** (def:inversion-handle). Defined object. *transported-score inversion handle*

Construction. For every  $\vartheta \in \Theta$ , form the source inverse-propensity score  $w(X)\{Z/e(X) - (1-Z)/(1-e(X))\}(Y - \vartheta D)$ , compare its empirical mean with  $L_\alpha\{n^{-1} \sum_{i=1}^n w(X_i)^2/n\}^{1/2}$ , and invert the resulting acceptance inequality over  $\Theta$ . Once  $w$  and  $e$  are oracle-known, the target sample does not enter this score.

Inputs.  $\mathcal{O}_i \sim \mathcal{S}; w; e$ .

**Definition 9** (def:geometry-handle). Defined object. *fixed-strength geometry handle*

Construction. Hold an arbitrary admissible deterministic array  $(P_S^X, w, e)$  fixed, set  $\mu_n = (t_0 \kappa_n/n)^{1/2}$  and the conditional complier probability  $p_n(x) = \mu_n w(x)/\kappa_n$ , vary only the complier outcome mean over a continuum, and compare every member with its center by chi-square and total variation.

Inputs.  $P_S \sim X; w; e; t_0$ .

**Definition 10** (def:least-favorable-witness). Defined object. *fixed-geometry continuum witness*

Construction. Fix  $\mathbf{g} = (P_S^X, w, e)_{n \geq 1} \in \mathcal{G}$  and  $t_0 > 0$ . Set  $\mu_n = (t_0 \kappa_n/n)^{1/2}$ ,  $p_n(x) = \mu_n w(x)/\kappa_n$ ,  $\rho = (1 - \alpha)/8$ , and  $H(t_0) = \min\{1/4, \rho t_0^{-1/2}\}$ . Conditional on  $X = x$ , assign compliance types  $(D(0), D(1)) = (0, 1), (1, 1), (0, 0)$  probabilities  $p_n(x), (1 - p_n(x))/2, (1 - p_n(x))/2$ . Set  $Y(0) = 0$ ; for compliers let  $Y(1)$  be Bernoulli  $(1/2 + h)$ , and for both always-takers and never-takers let  $Y(1)$  be Bernoulli  $(1/2)$ , for  $|h| \leq H(t_0)$ . Use this same conditional full-data law in both populations, draw source encouragement with propensity  $e(X)$ , and denote the resulting joint observed-data law and fixed oracle inputs by  $Q_{n,h}^{\mathbf{g}}$ .

Inputs.  $\backslash \text{mathfrak{g}}; t_0; \backslash \rho$ .

**Definition 11** (def:admissible-geometry-class). Defined object.  $\mathcal{G}$

Construction.  $\mathcal{G}$  is the class of deterministic arrays  $\mathbf{g} = (P_S^X, w, e)_{n \geq 1}$  such that  $P_S^X$  is a probability law on  $\mathcal{X}$ ,  $e : \mathcal{X} \rightarrow [\varepsilon, 1 - \varepsilon]$ ,  $w : \mathcal{X} \rightarrow [0, 2k_n]$ ,  $\mathbb{E}_S[w(X)] = 1$ ,  $\kappa_n = \mathbb{E}_S[w(X)^2] \leq k_n$ , and  $P_T(A) = \mathbb{E}_S[w(X)\mathbf{1}\{X \in A\}]$  for every measurable  $A \subseteq \mathcal{X}$ .

Inputs.  $P_S \sim X; w; e$ .

**Definition 12** (def:fixed-geometry-slice). Defined object.  $\mathcal{P}_n(\mathbf{g})$

Construction. For  $\mathbf{g} \in \mathcal{G}$ ,

$$\mathcal{P}_n(\mathbf{g}) = \{P \in \mathcal{P}_n : (P_S^X, w, e) \text{ equals the index-} n \text{ component of } \mathbf{g}\}$$

Inputs.  $\backslash \text{mathfrak{g}}$ .

**Definition 13** (def:fixed-geometry-honesty). Defined object. *fixed-geometry oracle-honest sequence*

Construction.

$$\mathfrak{C}^{\text{or}}(\mathbf{g}) = \{(C_n)_{n \geq 1} : C_n = C_n((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, w, e) \subseteq \Theta, \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n(\mathbf{g})} P(\theta_T \in C_n) \geq 1 - \alpha\}$$

Inputs.  $\mathcal{O}_i \sim \mathcal{S}; X_j \sim T; w; e$ .

**Definition 14** (def:fixed-geometry-value). Defined object.  $V_{\mathbf{g}}^*(t_0)$

Construction.

$$V_{\mathbf{g}}^*(t_0) = \inf_{(C_n) \in \mathfrak{C}^{\text{or}}(\mathbf{g})} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_n(\mathbf{g}) : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)]$$

**Definition 15** (def:regular-finite-cell-class). Defined object.  $\mathcal{N}_n^{\text{reg}}$

Construction.

$\mathcal{N}_n^{\text{reg}} = \{P \in \mathcal{P}_n : \mathcal{X} = \{1, \dots, k_n\}, q_{x,n} = P_S^X(X = x), c_-/k_n \leq q_{x,n} \leq c_+/k_n \text{ for every } x\}$ , where  $0 < c_- \leq 1 \leq c_+ < \infty$  are fixed.

**Definition 16** (def:regular-cell-feasible-honesty). Defined object. *regular-cell weight-unknown honest sequence*

Construction.

$\mathfrak{C}^{\text{reg}} = \{(C_n)_{n \geq 1} : C_n = C_n((O_i^S)_{i=1}^n, (X_j^T)_{j=1}^{N_n}, (q_{x,n})_{x=1}^{k_n}, e) \subseteq \Theta, \liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \in C_n) \geq 1 - \alpha\}$ .

Inputs.  $O_i^S; X_j^T; q_{x,n}; e$ .

## 6 Main results

**Proposition 1** (prop:compact-causal-range). For every  $P \in \mathcal{P}_n$ , source randomization identifies  $\Delta_Y(x) = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1]$  and  $\Delta_D(x) = \mathbb{E}_{P_n^F}[D(1) - D(0) \mid X = x, S = 1]$ . Outcome-contrast transport and integrated first-stage transport imply  $\mu_{Y,n} = \mathbb{E}_{P_n^F}[(Y(1) - Y(0))\mathbf{1}\{D(1) > D(0)\} \mid S = 0]$  and  $\mu_n = P_n^F(D(1) > D(0) \mid S = 0)$ . Hence  $\mu_{Y,n}/\mu_n = \theta_T = \mathbb{E}_{P_n^F}[Y(1) - Y(0) \mid D(1) > D(0), S = 0] \in \Theta = [-1, 1]$ .

*Proof.* Fix  $P \in \mathcal{P}_n$ . All conditional equalities below are understood for the relevant almost-everywhere versions. By source consistency and exclusion, for  $z \in \{0, 1\}$ ,

$$D = D(z) \quad \text{and} \quad Y = Y(D(z)) = Y^Z(z)$$

on the event  $\{Z = z, S = 1\}$ . Conditional instrument exchangeability implies that  $Z$  is conditionally independent, given  $(X, S = 1)$ , of both  $D(z)$  and  $Y^Z(z) = Y(D(z))$ , because the latter is a measurable function of  $(Y(0), Y(1), D(z))$ . Consequently,

$$\mathbb{E}_S[Y \mid Z = z, X = x] = \mathbb{E}_{P_n^F}[Y^Z(z) \mid X = x, S = 1]$$

and

$$\mathbb{E}_S[D \mid Z = z, X = x] = \mathbb{E}_{P_n^F}[D(z) \mid X = x, S = 1].$$

Subtracting the identities for  $z = 0$  from those for  $z = 1$ , and using the definitions of the observed source contrasts, gives

$$\Delta_Y(x) = \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X = x, S = 1]$$

and

$$\Delta_D(x) = \mathbb{E}_{P_n^F}[D(1) - D(0) \mid X = x, S = 1].$$

These equalities establish source identification of the two conditional contrasts.

Let  $A = \mathbf{1}\{D(1) > D(0)\}$ . Since  $D(0), D(1)$  are binary and monotonicity holds in both populations,

$$D(1) - D(0) = A.$$

Exclusion and the elementary binary-treatment identity  $Y(d) = Y(0) + d\{Y(1) - Y(0)\}$  further yield

$$Y^Z(1) - Y^Z(0) = Y(D(1)) - Y(D(0)) = \{D(1) - D(0)\}\{Y(1) - Y(0)\} = A\{Y(1) - Y(0)\}.$$

Transport domination and the definition  $w = dP_T/dP_S^X$  imply the change-of-measure identity  $\mathbb{E}_S[w(X)h(X)] = \mathbb{E}_T[h(X)]$  for every bounded measurable  $h$ . Applying it to  $h = \Delta_Y$ , then applying conditional outcome-contrast



transport and iterated expectation, gives

$$\begin{aligned}
\mu_{Y,n} &= \mathbb{E}_T[\Delta_Y(X)] \\
&= \mathbb{E}_T[\mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X, S = 1]] \\
&= \mathbb{E}_T[\mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid X, S = 0]] \\
&= \mathbb{E}_{P_n^F}[Y^Z(1) - Y^Z(0) \mid S = 0] \\
&= \mathbb{E}_{P_n^F}[\{Y(1) - Y(0)\}A \mid S = 0].
\end{aligned}$$

Similarly, change of measure, the identified receipt contrast, integrated first-stage transport, and iterated expectation imply

$$\begin{aligned}
\mu_n &= \mathbb{E}_T[\Delta_D(X)] \\
&= \mathbb{E}_{X \mid S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 1]] \\
&= \mathbb{E}_{X \mid S=0}[\mathbb{E}_{P_n^F}[D(1) - D(0) \mid X, S = 0]] \\
&= \mathbb{E}_{P_n^F}[D(1) - D(0) \mid S = 0] \\
&= P_n^F(D(1) > D(0) \mid S = 0).
\end{aligned}$$

Target complier positivity gives  $\mu_n > 0$ , so division is valid. Therefore

$$\frac{\mu_{Y,n}}{\mu_n} = \mathbb{E}_{P_n^F}[Y(1) - Y(0) \mid D(1) > D(0), S = 0] = \theta_T.$$

Finally, full-data support gives  $-1 \leq Y(1) - Y(0) \leq 1$  almost surely. A conditional expectation of this contrast also lies in  $[-1, 1]$ , and hence  $\theta_T \in [-1, 1] = \Theta$ , as claimed.  $\square$

*Justification.* The proposition makes the target a genuine causal estimand and establishes that compactness follows from the full-data outcome bound. *Gap.* Dufour (1997) addresses an unbounded locally unidentified coefficient, not an intrinsically bounded target complier effect. *Consumer.* It supplies the finite-diameter premise for the frontier.

**Theorem 1** (thm:oracle-converse). *There exist constants  $c_0 > 0$  and  $t_c > 0$ , depending only on  $\alpha, \varepsilon$ , and the fixed class constants, such that for every  $t_0 > 0$ ,  $V^*(t_0) \geq c_0 \min\{1, t_0^{-1/2}\}$  and  $V_N^*(t_0) \geq c_0 \min\{1, t_0^{-1/2}\}$ . In particular,  $V^*(t_0) \geq c_0$  whenever  $0 < t_0 \leq t_c$ . The constants do not depend on the decomposition of  $t_0$  into  $\mu_n$  and  $\kappa_n$ .*

*Proof.* Set  $c_0 = 3(1 - \alpha)^2/16$  and  $t_c = 1$ . Fix  $t_0 > 0$ . For any admissible deterministic geometry  $\mathbf{g}$ , every globally oracle-honest sequence is honest on the fixed-geometry slice  $\mathcal{P}_n(\mathbf{g})$ , and its global risk dominates its risk on that slice. Therefore the fixed-geometry frontier theorem gives

$$V^*(t_0) \geq V_{\mathbf{g}}^*(t_0) \geq c_0 \min\{1, t_0^{-1/2}\}.$$

For the finite-cell value, choose the admissible geometry for which  $P_S^X$  is uniform on  $\{1, \dots, k_n\}$ ,  $w = 1$ , and  $e = 1/2$ . Its fixed-geometry slice is contained in  $\mathcal{N}_n$ . By the no-shift reduction,  $\kappa_n = 1$  and the restriction  $t_n \geq t_0$  on this slice is exactly  $n\mu_n^2 \geq t_0$ . Every globally oracle-honest sequence is conditionally honest on the slice, and its finite-cell risk dominates the slice risk. Hence

$$V_{\mathcal{N}}^*(t_0) \geq V_{\mathbf{g}}^*(t_0) \geq c_0 \min\{1, t_0^{-1/2}\}.$$

These conclusions hold for every  $t_0 > 0$ . If  $0 < t_0 \leq t_c = 1$ , then the minimum equals one and thus  $V^*(t_0) \geq c_0$ . Finally, the fixed-geometry least-favourable construction holds  $P_S^X$ ,  $w$ ,  $e$ , and  $\kappa_n$  fixed and sets  $\mu_n = (t_0 \kappa_n / n)^{1/2}$ . Its total-variation calibration and therefore  $c_0$  depend on the pair  $(\mu_n, \kappa_n)$  only through  $t_0 = n\mu_n^2 / \kappa_n$ . This proves the asserted decomposition invariance.  $\square$

*Justification.* The converse tests whether Kish deflation is an information bound rather than a variance diagnostic. *Gap.* Mikusheva (2010) treats invariant linear-IV arc length, while Armstrong and Kolesar (2021) treat regular overlap-sensitive ATEs; neither supplies this transported causal-ratio converse. *Consumer.* It identifies regimes where even an oracle cannot give a short honest set in deep-canvassing and transported-encouragement studies.

**Proposition 2** (prop:no-shift-reduction). *If  $w(X) = 1$   $P_S$ -almost surely, then  $\kappa_n = 1$  and  $t_n = n\mu_n^2$ ; hence the frontier specializes to compact single-population weak-ratio inference.*

*Proof.* Assume  $w(X) = 1$   $P_S$ -almost surely. Since  $w$  is a function of  $X$ , this equality also holds  $P_S^X$ -almost surely. The definitions give

$$\kappa_n = \mathbb{E}_S[w(X)^2] = 1, \quad t_n = \frac{n\mu_n^2}{\kappa_n} = n\mu_n^2.$$

Moreover, because  $w = dP_T/dP_S^X$ , every measurable  $A \subseteq \mathcal{X}$  satisfies

$$P_T(A) = \int_A w(x) dP_S^X(x) = P_S^X(A),$$

so there is no covariate-distribution shift. The compact-range proposition gives the forced parameter space  $\Theta = [-1, 1]$ . Now fix any admissible no-shift geometry  $\mathbf{g}$  with  $w \equiv 1$ . For every fixed  $t_0 > 0$ , the fixed-geometry frontier theorem gives the lower bound

$$V_{\mathbf{g}}^*(t_0) \geq \frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\},$$

whereas the oracle score-inversion rule applies on this slice and gives the matching upper order. Since  $t_n = n\mu_n^2$ , the strength restriction is precisely  $n\mu_n^2 \geq t_0$ . Thus the no-shift conditional frontier is the compact single-population weak-ratio frontier, up to the universal constants in the two theorems, and it is indexed by  $n\mu_n^2$ . The dependency order is acyclic: the fixed-geometry theorem supplies the lower bound, the score-inversion theorem supplies the upper bound, and neither conclusion is obtained from the global converse.  $\square$

*Justification.* When  $w = 1$ , the effective-strength index reduces to the familiar single-population weak-ratio index  $n\mu_n^2$ . *Gap.* The proposition is an index reduction; the fixed-geometry theorem supplies the separate two-sided minimax statement on the no-shift slice. *Consumer.* It connects the transported frontier to compact single-population weak-ratio inference without conflating the two experiments.

**Theorem 2** (thm:oracle-score-inversion-attainment). *Let*

$$H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}, \quad \hat{A}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i Y_i, \quad \hat{B}_n = \frac{1}{n} \sum_{i=1}^n w(X_i) H_i D_i,$$

and  $\hat{\kappa}_n = n^{-1} \sum_{i=1}^n w(X_i)^2$ . With  $L_\alpha = \{8/(\alpha\epsilon^2)\}^{1/2}$ , define the single score-inversion sequence

$$C_n = \left\{ \vartheta \in \Theta : |\hat{A}_n - \vartheta \hat{B}_n| \leq L_\alpha \sqrt{\hat{\kappa}_n / n} \right\}.$$

Then  $(C_n) \in \mathfrak{C}^{\text{or}}$ , and, simultaneously for every  $t_0 > 0$ ,

$$R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}, \quad C_0 = \max\left\{2, 4L_\alpha + \frac{8}{\varepsilon^2}\right\}.$$

Consequently  $V^*(t_0)$  has order  $\min\{1, t_0^{-1/2}\}$ . The displayed upper bound holds for every admissible weight geometry and depends on a member law through  $t_n = n\mu_n^2/\kappa_n$  only, apart from the fixed constants. Combined with the fixed-geometry frontier theorem, the achievability bound and the conditional converse match in order uniformly over admissible geometries, so any two admissible geometries with the same effective strength are equally difficult up to universal constants.

*Proof.* For  $\vartheta \in \Theta$ , define

$$\psi_{i,\vartheta} = w(X_i)H_i(Y_i - \vartheta D_i).$$

Conditional randomization, consistency, and the definition of  $H_i$  imply

$$\mathbb{E}_P[\psi_{i,\vartheta}] = \mu_{Y,n} - \vartheta \mu_n.$$

By the compact-range proposition this mean is zero at  $\vartheta = \theta_T$ . Instrument overlap and boundedness give  $|H_i| \leq \varepsilon^{-1}$  and  $|Y_i - \theta_T D_i| \leq 2$ , whence

$$\text{Var}_P(\psi_{i,\theta_T}) \leq \frac{4\kappa_n}{\varepsilon^2}.$$

The weight envelope gives  $w^4 \leq 4k_n^2 w^2$ , so

$$\text{Var}_P(\widehat{\kappa}_n) \leq \frac{\mathbb{E}_P[w(X)^4]}{n} \leq \frac{4k_n^2 \kappa_n}{n}.$$

Since  $\mathbb{E}_P[w(X)] = 1$ , Cauchy–Schwarz gives  $\kappa_n \geq 1$ . Because  $k_n = o(n^{1/2})$  and  $\mathbb{E}_P[\widehat{\kappa}_n] = \kappa_n$ , Chebyshev’s inequality yields, uniformly over  $P \in \mathcal{P}_n$ ,

$$P(\widehat{\kappa}_n < \kappa_n/2) \leq \frac{16k_n^2}{n\kappa_n} = o(1).$$

On the complementary event, failure to include  $\theta_T$  implies

$$\left| \frac{1}{n} \sum_{i=1}^n \psi_{i,\theta_T} \right| > L_\alpha \sqrt{\frac{\kappa_n}{2n}}.$$

A second application of Chebyshev’s inequality gives

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n \psi_{i,\theta_T}\right| > L_\alpha \sqrt{\frac{\kappa_n}{2n}}\right) \leq \frac{8}{L_\alpha^2 \varepsilon^2} = \alpha.$$

Consequently

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_n} P(\theta_T \in C_n) \geq 1 - \alpha,$$

so  $(C_n) \in \mathfrak{C}^{\text{or}}$ .

For the length bound, put  $r_n = L_\alpha \sqrt{\widehat{\kappa}_n/n}$ . The acceptance set is the intersection of  $\Theta$  with the inverse image of  $[-r_n, r_n]$  under the affine map  $\vartheta \mapsto \widehat{A}_n - \vartheta \widehat{B}_n$ . Therefore

$$\lambda(C_n) \leq \begin{cases} \min\{2, 2r_n/|\widehat{B}_n|\}, & \widehat{B}_n \neq 0, \\ 2, & \widehat{B}_n = 0. \end{cases}$$

The summand  $w(X_i)H_iD_i$  has mean  $\mu_n$  and variance at most  $\kappa_n/\varepsilon^2$ . Thus

$$P(|\widehat{B}_n - \mu_n| > \mu_n/2) \leq \frac{4\kappa_n}{\varepsilon^2 n \mu_n^2} = \frac{4}{\varepsilon^2 t_n}.$$

On the complementary event,  $|\widehat{B}_n| \geq \mu_n/2$ , while Jensen's inequality gives

$$\mathbb{E}_P[r_n] \leq L_\alpha \sqrt{\frac{\mathbb{E}_P[\widehat{\kappa}_n]}{n}} = L_\alpha \sqrt{\frac{\kappa_n}{n}}.$$

It follows that

$$\mathbb{E}_P[\lambda(C_n)] \leq \frac{4L_\alpha}{\sqrt{t_n}} + \frac{8}{\varepsilon^2 t_n}.$$

For  $t_0 \geq 1$ , taking the supremum over  $t_n \geq t_0$  bounds the right-hand side by  $(4L_\alpha + 8\varepsilon^{-2})t_0^{-1/2}$ . For  $0 < t_0 < 1$ , use  $\lambda(C_n) \leq 2$ . Taking the required limit superior proves

$$R((C_n), t_0) \leq C_0 \min\{1, t_0^{-1/2}\}.$$

For the matching lower order, fix any  $\mathbf{g} \in \mathcal{G}$ . Every globally oracle-honest sequence is honest on  $\mathcal{P}_n(\mathbf{g})$ , and its global risk dominates its fixed-slice risk. Hence the lower half of the fixed-geometry frontier theorem implies

$$V^\star(t_0) \geq V_{\mathbf{g}}^\star(t_0) \geq \frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

Together with the displayed achievable upper bound, this proves the global minimax order. The upper-bound calculation uses a geometry only through  $\kappa_n$ , while the fixed-geometry theorem supplies a matching conditional converse with constants uniform in  $\mathbf{g}$ . Thus the two sides match uniformly over admissible geometries, and geometries with the same effective strength are equally difficult up to universal constants.  $\square$

**Theorem 3** (thm:finite-cell-unknown-weight-attainment). *On  $\mathcal{N}_n$ , write*

$$H_i = 2(2Z_i - 1), \quad \widehat{p}_{x,n} = \frac{1}{N_n} \sum_{j=1}^{N_n} \mathbf{1}\{X_j^T = x\},$$

and, for  $G \in \{Y, D\}$ , define

$$\widehat{m}_{G,n}(x) = \frac{k_n}{n} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i, \quad \widehat{M}_{G,n} = \sum_{x=1}^{k_n} \widehat{p}_{x,n} \widehat{m}_{G,n}(x).$$

For  $N_n \geq 2$ , let

$$\widehat{\kappa}_{U,n} = \frac{k_n}{N_n(N_n - 1)} \sum_{j \neq \ell} \mathbf{1}\{X_j^T = X_\ell^T\}, \quad \widehat{K}_n = 1 + \widehat{\kappa}_{U,n}.$$

Put  $B_c = 32(1 + c^{-1})$ ,  $L_{\alpha,c} = \{2B_c/\alpha\}^{1/2}$ , and

$$C_n = \left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L_{\alpha,c} \sqrt{\widehat{K}_n/n} \right\},$$

with  $C_n = \Theta$  at the finitely many indices for which  $N_n < 2$ . This sequence belongs to  $\mathfrak{C}^{\text{cell}}$ , does not use  $w$ , and satisfies, simultaneously for all  $t_0 > 0$ ,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \leq C_0 \min\{1, t_0^{-1/2}\},$$

where

$$C_0 = \max\{2, 4\sqrt{2}L_{\alpha,c} + 8B_c\}.$$

Moreover, the finite-cell change-of-measure family used for the oracle converse applies to every sequence in  $\mathfrak{C}^{\text{cell}}$ ; hence the corresponding infimum of the displayed worst-case risk is at least  $c_0 \min\{1, t_0^{-1/2}\}$ . Thus estimating the target cell probabilities creates no additional minimax-order loss on  $\mathcal{N}_n$ .

*Proof.* Specialize the regular-cell theorem to

$$q_{x,n} = \frac{1}{k_n}, \quad e(x) = \frac{1}{2}, \quad L = L_{\alpha,c} = \left\{ \frac{2B_c}{\alpha} \right\}^{1/2}, \quad B_c = 32(1 + c^{-1}).$$

Then its score is  $H_i = 2(2Z_i - 1)$ , and its source-cell estimator becomes

$$\frac{1}{nq_{x,n}} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i = \frac{k_n}{n} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i.$$

Its collision estimator likewise becomes

$$\frac{1}{N_n(N_n - 1)} \sum_{j \neq \ell} \frac{\mathbf{1}\{X_j^T = X_\ell^T\}}{q_{X_j^T, n}} = \frac{k_n}{N_n(N_n - 1)} \sum_{j \neq \ell} \mathbf{1}\{X_j^T = X_\ell^T\}.$$

Because  $c > 0$ , one has  $(4 + 3/c)^{-1/2} < 1/2$ . Choose

$$\bar{\varepsilon} = \frac{1}{4} + \frac{1}{2}(4 + 3/c)^{-1/2}.$$

Then  $\bar{\varepsilon} \in (0, 1/2)$  and  $\bar{\varepsilon}^{-2} \leq 4 + 3/c$ . Since  $e(x) = 1/2$  on  $\mathcal{N}_n$ ,

$$\bar{\varepsilon} \leq e(x) \leq 1 - \bar{\varepsilon},$$

so the regular-cell theorem applies with the admissible interior overlap constant  $\bar{\varepsilon}$ . Its variance constant satisfies

$$\begin{aligned} B_{\bar{\varepsilon}, c} &= 8(\bar{\varepsilon}^{-2} + c^{-1}) \\ &\leq 8(4 + 3c^{-1} + c^{-1}) = 32(1 + c^{-1}) = B_c. \end{aligned}$$

Therefore  $L_{\alpha,c} \geq \{2B_{\bar{\varepsilon}, c}/\alpha\}^{1/2}$ , so the exact confidence set in the statement is honest. Because the specialized construction uses only the two observed samples, it lies in  $\mathfrak{C}^{\text{cell}}$  and does not use  $w$ . The regular-cell length bound gives

$$\begin{aligned} &\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \\ &\leq \max\{2, 4\sqrt{2}L_{\alpha,c} + 8B_{\bar{\varepsilon}, c}\} \min\{1, t_0^{-1/2}\} \\ &\leq C_0 \min\{1, t_0^{-1/2}\}. \end{aligned}$$

For the converse, use the fixed geometry  $q_{x,n} = 1/k_n$ ,  $w = 1$ , and  $e = 1/2$ . It is a submodel of  $\mathcal{N}_n$ , and the target-sample law and all known design inputs are fixed throughout the continuum of outcome alternatives in

the fixed-geometry construction. Every sequence in  $\mathfrak{C}^{\text{cell}}$  is therefore conditionally honest on this slice, while its finite-cell risk dominates its slice risk. The lower half of the fixed-geometry frontier theorem, equivalently the finite-cell specialization used in the oracle converse, yields

$$\inf_{(C_n) \in \mathfrak{C}^{\text{cell}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \geq c_0 \min\{1, t_0^{-1/2}\},$$

with  $c_0 = 3(1 - \alpha)^2/16$ . The upper and lower orders coincide, proving that estimating the target cell probabilities causes no additional minimax-order loss.  $\square$

*Justification.* The result separates learning the transport weights from weak-ratio identification: in the stated uniform-cell experiment, the target covariate sample can learn the cell probabilities without changing the honest expected-length order. *Gap.* The closest transported-CACE reference, Chen and Huang (2025, [1]), supplies the transported identifying ratio and regular weighted or multiply robust asymptotics; it does not analyze a covariate-only target sample that learns unknown target cell probabilities, nor worst-case honest expected set length as the target first stage vanishes. Choi, Gu, and Shen (2018, [5]) give weak-IV-robust procedures for linear two-sample IV, not a transport-weight learning problem with a bounded causal-ratio target, known uniform source cells, and a collision scale estimate. Thus neither comparator establishes the no-additional-order-loss conclusion for  $\mathcal{N}_n$ . *Consumer.* It identifies a genuinely feasible benchmark against the oracle frontier and provides the uniform-cell specialization consumed by the regular-cell extension.

**Theorem 4** (thm:fixed-geometry-frontier). *Let  $\mathbf{g} \in \mathcal{G}$  be any admissible deterministic geometry array. Put  $L_\alpha = \{8/(\alpha\varepsilon^2)\}^{1/2}$ ,  $c_\alpha = 3(1 - \alpha)^2/16$ , and  $C_\alpha = \max\{2, 4L_\alpha + 8\varepsilon^{-2}\}$ . Then, for every fixed  $t_0 > 0$ ,*

$$c_\alpha \min\{1, t_0^{-1/2}\} \leq V_{\mathbf{g}}^*(t_0) \leq C_\alpha \min\{1, t_0^{-1/2}\}.$$

*Both constants are uniform in  $\mathbf{g}$ . In particular, after conditioning on any admissible source-covariate law, transport-weight array, and propensity array, geometries having the same effective strength are equally difficult up to the displayed universal constants.*

*Proof.* Fix  $\mathbf{g} \in \mathcal{G}$  and  $t_0 > 0$ . We prove the lower and upper bounds separately.

For the lower bound, abbreviate the fixed geometry by  $(P_S^X, w_n, e_n)$ , let  $\kappa_n = \mathbb{E}_S[w_n(X)^2]$ , and set

$$\mu_n = \sqrt{\frac{t_0 \kappa_n}{n}}, \quad p_n(x) = \frac{\mu_n w_n(x)}{\kappa_n}.$$

The normalization  $\mathbb{E}_S[w_n(X)] = 1$  implies  $\kappa_n \geq 1$ . The envelope and the degrading-array condition then give

$$0 \leq \sup_x p_n(x) \leq \frac{2\sqrt{t_0} k_n}{\sqrt{n\kappa_n}} \leq \frac{2\sqrt{t_0} k_n}{\sqrt{n}} = o(1),$$

so  $p_n(x)$  is a valid compliance probability for all sufficiently large  $n$ . Also

$$\mu_n^2 \leq \frac{t_0 k_n}{n} \longrightarrow 0,$$

so the constructed triangular array satisfies  $\mu_n \rightarrow 0$ .

Use the fixed-geometry continuum witness: conditional on  $X = x$ , assign complier, always-taker, and never-taker probabilities  $p_n(x)$ ,  $(1 - p_n(x))/2$ , and  $(1 - p_n(x))/2$ . Set  $Y(0) = 0$ . For compliers let  $Y(1)$  be Bernoulli with success probability  $1/2 + h$ ; for noncompliers use success probability  $1/2$  for  $Y(1)$ . Use the same conditional full-data law in both populations and draw source encouragement with propensity  $e_n(X)$ . This construction has no defiers, satisfies exclusion and conditional randomization, and, because the conditional full-data laws agree between populations, satisfies both transport restrictions. Its target complier share is

$$\mathbb{E}_T[p_n(X)] = \mathbb{E}_S[w_n(X)p_n(X)] = \frac{\mu_n}{\kappa_n} \mathbb{E}_S[w_n(X)^2] = \mu_n > 0.$$

Consequently

$$t_n = \frac{n\mu_n^2}{\kappa_n} = t_0, \quad \theta_T = \frac{1}{2} + h.$$

All laws in the continuum have exactly the same source covariate law, transport weight, target covariate law, propensity, Kish dispersion, and effective strength.

Take

$$\rho = \frac{1-\alpha}{8}, \quad H(t_0) = \min \left\{ \frac{1}{4}, \frac{\rho}{\sqrt{t_0}} \right\},$$

and restrict  $|h| \leq H(t_0)$ . Conditional on  $X = x$ , the parameter  $h$  changes the observed source law only on  $\{Z = 1, D = 1\}$ . Under the central law  $h = 0$ , this event has probability  $e_n(x)(1 + p_n(x))/2$ ; within it, the complier fraction is  $2p_n(x)/(1 + p_n(x))$ , and hence  $Y$  is Bernoulli with success probability

$$\frac{1}{2} + \frac{2p_n(x)h}{1 + p_n(x)}.$$

The chi-square divergence of Bernoulli  $(1/2 + u)$  from Bernoulli  $(1/2)$  is  $4u^2$ . Therefore the one-observation conditional contribution is

$$\frac{e_n(x)(1 + p_n(x))}{2} 4 \left\{ \frac{2p_n(x)h}{1 + p_n(x)} \right\}^2 = \frac{8e_n(x)p_n(x)^2h^2}{1 + p_n(x)} \leq 8p_n(x)^2h^2.$$

Averaging over  $P_S^X$ , using  $\mathbb{E}_S[p_n(X)^2] = \mu_n^2/\kappa_n$ , and taking the product of the  $n$  independent source laws gives

$$1 + \chi^2(Q_{n,h}^{\mathfrak{g}} \| Q_{n,0}^{\mathfrak{g}}) \leq \left( 1 + \frac{8\mu_n^2h^2}{\kappa_n} \right)^n \leq \exp(8t_0h^2) \leq \exp(8\rho^2).$$

The target sample and oracle inputs have the same distribution for every  $h$ . For a likelihood ratio  $L$ , Cauchy–Schwarz gives  $\|Q - Q_0\|_{\text{TV}} \leq \frac{1}{2}\{\mathbb{E}_{Q_0}(L - 1)^2\}^{1/2}$ . Since  $8\rho^2 \leq 1$  and  $e^u - 1 \leq 2u$  for  $0 \leq u \leq 1$ ,

$$\sup_{|h| \leq H(t_0)} \|Q_{n,h}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}} \leq 2\rho = \frac{1-\alpha}{4}.$$

Let  $(C_n) \in \mathfrak{C}^{\text{or}}(\mathfrak{g})$ . Conditional honesty supplies a deterministic  $\gamma_n \downarrow 0$  such that coverage throughout the slice is at least  $1 - \alpha - \gamma_n$  for all sufficiently large  $n$ . Put  $I = [1/2 - H(t_0), 1/2 + H(t_0)]$ . Tonelli's theorem and the total-variation inequality imply

$$\begin{aligned} \mathbb{E}_{Q_{n,0}^{\mathfrak{g}}}[\lambda(C_n)] &= \int_{\Theta} Q_{n,0}^{\mathfrak{g}}(u \in C_n) du \\ &\geq \int_I \left\{ Q_{n,u-1/2}^{\mathfrak{g}}(u \in C_n) - \|Q_{n,u-1/2}^{\mathfrak{g}} - Q_{n,0}^{\mathfrak{g}}\|_{\text{TV}} \right\} du \\ &\geq 2H(t_0) \left\{ \frac{3(1-\alpha)}{4} - \gamma_n \right\}. \end{aligned}$$

Because  $\rho \leq 1/4$ ,

$$H(t_0) \geq \rho \min\{1, t_0^{-1/2}\}.$$

Taking limits and then the infimum over conditionally honest sequences gives

$$V_{\mathfrak{g}}^*(t_0) \geq \frac{3(1-\alpha)\rho}{2} \min\{1, t_0^{-1/2}\} = \frac{3(1-\alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

For the upper bound, use the source-score inversion rule

$$C_n^{\text{sc}} = \left\{ \vartheta \in \Theta : \left| \frac{1}{n} \sum_{i=1}^n w_n(X_i) \left\{ \frac{Z_i}{e_n(X_i)} - \frac{1 - Z_i}{1 - e_n(X_i)} \right\} (Y_i - \vartheta D_i) \right| \leq L_\alpha \sqrt{\frac{\hat{\kappa}_n}{n}} \right\},$$

where  $\hat{\kappa}_n = n^{-1} \sum_i w_n(X_i)^2$ . At  $\vartheta = \theta_T$ , the score has mean zero by the compact-range proposition and variance at most  $4\kappa_n/\varepsilon^2$ . The envelope gives

$$\text{Var}(\hat{\kappa}_n) \leq \frac{4k_n^2 \kappa_n}{n}, \quad \sup_{P \in \mathcal{P}_n(\mathfrak{g})} P(\hat{\kappa}_n < \kappa_n/2) \leq \frac{16k_n^2}{n\kappa_n} = o(1).$$

Chebyshev's inequality therefore bounds noncoverage by

$$\frac{8}{L_\alpha^2 \varepsilon^2} + o(1) = \alpha + o(1),$$

uniformly on the slice, so the rule is conditionally honest.

Writing its radius as  $r_n = L_\alpha \sqrt{\hat{\kappa}_n/n}$ , affine inversion gives length at most  $2r_n/|\hat{B}_n|$ , truncated at two, where

$$\hat{B}_n = \frac{1}{n} \sum_{i=1}^n w_n(X_i) \left\{ \frac{Z_i}{e_n(X_i)} - \frac{1 - Z_i}{1 - e_n(X_i)} \right\} D_i.$$

This denominator has mean  $\mu_n$  and variance at most  $\kappa_n/(n\varepsilon^2)$ . Hence

$$P(|\hat{B}_n - \mu_n| > \mu_n/2) \leq \frac{4}{\varepsilon^2 t_n}, \quad \mathbb{E}[r_n] \leq L_\alpha \sqrt{\frac{\kappa_n}{n}},$$

and consequently

$$\mathbb{E}_P[\lambda(C_n^{\text{sc}})] \leq \frac{4L_\alpha}{\sqrt{t_n}} + \frac{8}{\varepsilon^2 t_n}.$$

For  $t_0 \geq 1$ , take the supremum over  $t_n \geq t_0$ ; for  $t_0 < 1$ , use  $\lambda(C_n^{\text{sc}}) \leq 2$ . This gives

$$V_{\mathfrak{g}}^*(t_0) \leq \max\{2, 4L_\alpha + 8\varepsilon^{-2}\} \min\{1, t_0^{-1/2}\}.$$

All constants are independent of  $\mathfrak{g}$ , proving both the two-sided bound and the asserted fixed-geometry invariance.  $\square$

*Justification.* This is the theorem-level boundary: it turns the transported CACE ratio into a conditional minimax expected-length experiment, holding the entire source-covariate, transport-weight, and propensity geometry fixed while allowing the first stage to vanish and the Kish dispersion to grow. Its novelty claim is only this joint scope, not a claim to subsume regular transport, linear two-sample weak-IV, or discrete-ATE minimax theory. *Gap.* Chen and Huang (2025, [1]) identify the transported CACE Wald ratio and give regular weighted and multiply robust asymptotics, but do not give a fixed-geometry honest expected-length minimax frontier under  $\mu_n \rightarrow 0$  and  $\kappa_n \rightarrow \infty$ . Zeng, Kennedy, Bodnar, and Naimi (2024, [10]) give regular transport/generalization efficiency and minimax results for target causal effects, rather than a noncompliance ratio with a vanishing first stage or confidence-set length conditional on a fixed geometry. Choi and Shen (2019, [6]) give weak-instrument-robust confidence sets for a linear two-sample IV model, but not a covariate-only target transport experiment, density-ratio learning, or this conditional minimax class. The accepted-bank result `stat_discrete_ate_minimax_loggap` instead treats MSE for a discrete unconfounded ATE with a dimension-overlap frontier; it has neither an IV ratio nor transport/Kish geometry nor an honest expected-length criterion. Mikusheva (2010, [7]) studies near-shortest expected arc length in an invariant linear-IV experiment, not a bounded transported CACE conditional on an arbitrary deterministic weight geometry. The new boundary is therefore limited to uniform, geometry-by-geometry order equivalence indexed only by  $n\mu_n^2/\kappa_n$ . *Consumer.* It supplies the conditional lower bound used by the global oracle converse and certifies that the oracle score bound is not hiding a favorable transport geometry.



**Theorem 5** (thm:regular-cell-unknown-weight-attainment). On  $\mathcal{N}_n^{\text{reg}}$ , let  $q_{x,n} = P_S^X(X = x)$ , let

$$H_i = \frac{Z_i}{e(X_i)} - \frac{1 - Z_i}{1 - e(X_i)}, \quad \hat{p}_{x,n} = \frac{1}{N_n} \sum_{j=1}^{N_n} \mathbf{1}\{X_j^T = x\},$$

and, for  $G \in \{Y, D\}$ , define

$$\hat{m}_{G,n}(x) = \frac{1}{nq_{x,n}} \sum_{i=1}^n \mathbf{1}\{X_i = x\} H_i G_i, \quad \widehat{M}_{G,n} = \sum_{x=1}^{k_n} \hat{p}_{x,n} \hat{m}_{G,n}(x).$$

For  $N_n \geq 2$ , set

$$\hat{\kappa}_{U,n} = \frac{1}{N_n(N_n - 1)} \sum_{j \neq \ell} \frac{\mathbf{1}\{X_j^T = X_\ell^T\}}{q_{X_j^T,n}}, \quad \widehat{K}_n = 1 + \hat{\kappa}_{U,n}.$$

Put  $B_{\varepsilon,c} = 8(\varepsilon^{-2} + c^{-1})$ . For every fixed  $L \geq \{2B_{\varepsilon,c}/\alpha\}^{1/2}$ , the sequence

$$C_n^{\text{reg}} = \left\{ \vartheta \in \Theta : |\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n}| \leq L \sqrt{\widehat{K}_n/n} \right\},$$

with  $C_n^{\text{reg}} = \Theta$  when  $N_n < 2$ , belongs to  $\mathfrak{C}^{\text{reg}}$ , does not use  $w$ , and satisfies, for every fixed  $t_0 > 0$ ,

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}} : t_n \geq t_0} \mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \max\{2, 4\sqrt{2}L + 8B_{\varepsilon,c}\} \min\{1, t_0^{-1/2}\}.$$

The infimum of this risk over  $\mathfrak{C}^{\text{reg}}$  is at least  $3(1 - \alpha)^2 \min\{1, t_0^{-1/2}\}/16$ . Thus unknown target cell probabilities cause no minimax-order loss for known regular source-cell probabilities and a known cell-varying propensity bounded by  $\varepsilon$ .

*Proof.* Fix  $P \in \mathcal{N}_n^{\text{reg}}$ , write  $p_x = P_T(X = x)$ , and observe that

$$w(x) = \frac{p_x}{q_{x,n}}, \quad \kappa_n = \sum_{x=1}^{k_n} \frac{p_x^2}{q_{x,n}}.$$

For  $\vartheta \in \Theta$ , set

$$g_{i,\vartheta} = H_i(Y_i - \vartheta D_i), \quad a_\vartheta(x) = \mathbb{E}_S[g_{i,\vartheta} \mid X_i = x].$$

Conditional randomization gives

$$a_\vartheta(x) = \Delta_Y(x) - \vartheta \Delta_D(x), \quad \sum_x p_x a_{\theta_T}(x) = \mu_{Y,n} - \theta_T \mu_n = 0.$$

Moreover,  $|g_{i,\vartheta}| \leq 2/\varepsilon$  and  $|a_\vartheta(x)| \leq 2$ . The estimated transported moment is the cross-average

$$\widehat{M}_{Y,n} - \vartheta \widehat{M}_{D,n} = \frac{1}{n} \sum_{i=1}^n \frac{\hat{p}_{X_i,n}}{q_{X_i,n}} g_{i,\vartheta}.$$

Conditional on the target sample, its source-sample mean is  $\sum_x \hat{p}_{x,n} a_{\vartheta}(x)$ , and its conditional variance is at most

$$\frac{4}{\varepsilon^2 n} \sum_{x=1}^{k_n} \frac{\hat{p}_{x,n}^2}{q_{x,n}}.$$

At  $\vartheta = \theta_T$ , independence of the samples and the zero target mean yield

$$\text{Var}_P \left( \sum_x \hat{p}_{x,n} a_{\theta_T}(x) \right) \leq \frac{4}{N_n}.$$

The multinomial second-moment identity gives

$$\begin{aligned} \mathbb{E}_P \left[ \sum_x \frac{\hat{p}_{x,n}^2}{q_{x,n}} \right] &= \kappa_n + \frac{1}{N_n} \sum_x \frac{p_x - p_x^2}{q_{x,n}} \\ &\leq \kappa_n + \frac{k_n}{c_- N_n}. \end{aligned}$$

Since  $\kappa_n \geq 1$ ,  $N_n/n \rightarrow c$ , and  $k_n/N_n \rightarrow 0$ , the law of total variance gives, uniformly over the class for all sufficiently large  $n$ ,

$$\text{Var}_P(\widehat{M}_{Y,n} - \theta_T \widehat{M}_{D,n}) \leq B_{\varepsilon,c} \frac{\kappa_n}{n}, \quad B_{\varepsilon,c} = 8(\varepsilon^{-2} + c^{-1}).$$

The same bound applies to  $\widehat{M}_{D,n}$ , whose expectation is  $\mu_n$ .

The collision component is unbiased:

$$\mathbb{E}_P[\widehat{\kappa}_{U,n}] = \sum_x \frac{p_x^2}{q_{x,n}} = \kappa_n.$$

Its symmetric kernel is bounded by  $k_n/c_-$ . In the variance expansion, disjoint index pairs are independent; bounding the order  $N_n^2$  individual variance terms and order  $N_n^3$  overlapping-pair covariance terms by the squared kernel envelope gives

$$\text{Var}_P(\widehat{\kappa}_{U,n}) \leq \frac{32k_n^2}{c_-^2 N_n}.$$

If  $\kappa_n \leq 2$ , then  $\widehat{K}_n \geq 1 \geq \kappa_n/2$  deterministically. If  $\kappa_n > 2$ , Chebyshev's inequality gives

$$P(\widehat{K}_n < \kappa_n/2) \leq \frac{128k_n^2}{c_-^2 N_n \kappa_n^2} \leq \frac{128k_n^2}{c_-^2 N_n} = o(1),$$

uniformly, because  $k_n = o(n^{1/2})$  and  $N_n/n \rightarrow c$ .

On  $\{\widehat{K}_n \geq \kappa_n/2\}$ , exclusion of  $\theta_T$  implies that its centered moment exceeds  $L\sqrt{\kappa_n/(2n)}$ . Thus

$$\sup_{P \in \mathcal{N}_n^{\text{reg}}} P(\theta_T \notin C_n^{\text{reg}}) \leq o(1) + \frac{2B_{\varepsilon,c}}{L^2} \leq \alpha + o(1).$$

This proves  $(C_n^{\text{reg}}) \in \mathfrak{C}^{\text{reg}}$ .

For length, let  $r_n = L\sqrt{\hat{K}_n/n}$ . Affine inversion gives length at most  $4r_n/\mu_n$  on  $\{|\hat{M}_{D,n} - \mu_n| \leq \mu_n/2\}$ , and at most two otherwise. Since

$$P(|\hat{M}_{D,n} - \mu_n| > \mu_n/2) \leq \frac{4B_{\varepsilon,c}}{t_n}, \quad \mathbb{E}_P[\hat{K}_n] = 1 + \kappa_n \leq 2\kappa_n,$$

Jensen's inequality yields

$$\mathbb{E}_P[\lambda(C_n^{\text{reg}})] \leq \frac{4\sqrt{2}L}{\sqrt{t_n}} + \frac{8B_{\varepsilon,c}}{t_n}.$$

Taking the supremum over  $t_n \geq t_0$  for  $t_0 \geq 1$ , and using  $\lambda(C_n^{\text{reg}}) \leq 2$  for  $t_0 < 1$ , proves the stated upper bound.

For the converse, take the regular geometry  $q_{x,n} = 1/k_n$ ,  $w = 1$ , and  $e = 1/2$ . It lies in  $\mathcal{N}_n^{\text{reg}}$  because  $c_- \leq 1 \leq c_+$ . Every regular-cell-honest sequence is conditionally honest on this fixed geometry, while its regular-class risk dominates its slice risk. The fixed-geometry frontier theorem therefore gives

$$\inf_{(C_n) \in \mathfrak{C}^{\text{reg}}} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{N}_n^{\text{reg}}: t_n \geq t_0} \mathbb{E}_P[\lambda(C_n)] \geq \frac{3(1 - \alpha)^2}{16} \min\{1, t_0^{-1/2}\}.$$

This completes the matched-order result.  $\square$

## 7 Interpretation

The order-one elbow may be normalized at  $t_c = 1$ , but this is not a sharp critical constant. Below an order-one effective-strength level the theorem proves a nonvanishing lower bound on worst-case expected length; it does not claim that every honest set converges to the full diameter two. The same score-inversion sequence obeys the bound at each fixed  $t_0 > 0$ ; this is not a joint moderate-deviation assertion for moving thresholds  $t_{0,n}$ .

## Technical internal limitation (diagnostic only)

The order result deliberately uses crude Chebyshev and collision-U-statistic constants and does not claim practical or sharp calibration. The fixed-geometry converse is complete. The remaining technical limitation is feasibility outside known regular finite-cell source designs: unknown propensities require control of random cell denominators and their scale estimate, while continuous-covariate sieves require approximation assumptions, weak-denominator bias control, and a partition-complexity rule.

## 8 Honest scope

Proved results are the transported identifying ratio, its forced compact range, the matched global and fixed-geometry oracle frontiers, one geometry-uniform oracle score-inversion sequence, and matched feasible target-cell weight learning on the uniform class  $\mathcal{N}_n$  and a known regular nonuniform-cell extension with known cell-varying propensity. The oracle result is not presented as a feasible general-model procedure. No claim is made for unknown assignment propensities, continuous-covariate weight learning, sieve approximation, moving strength thresholds, disconnected-set taxonomy, or sharp constants.

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